SPORTS & CO-CURRICULAR MODULE (STUDENT ENGAGEMENT & TALENT DEVELOPMENT ENGINE)

SmartCampus School Management System
Technical Specification & Implementation Guide — v1.0 • March 2026
Confidential • Internal Documentation

Audience: School Administrators • Teachers • Coaches • Students • Parents
Related Modules: Student Management • Timetable • Academics • Attendance • Notifications • Parent Portal • Analytics

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1. MODULE OVERVIEW

1.1 Purpose

The Sports & Co-Curricular Module manages all non-academic student activities, including sports, clubs, arts, and competitions. It enables:

- Management of clubs, teams, and activities
- Student participation and attendance tracking
- Scheduling of events and competitions
- Performance tracking and awards management
- Communication between students, teachers, and parents

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2. CORE OBJECTIVES

1. Provide centralized management of co-curricular activities
2. Track student participation and engagement
3. Monitor performance in sports, arts, and competitions
4. Enable scheduling and coordination of events
5. Provide recognition and motivation through awards
6. Integrate co-curricular data with academic and student profiles
7. Deliver analytics for participation and performance trends
8. Support communication and notifications

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3. REAL-WORLD SYSTEM FLOW

3.1 Lifecycle Flow

Club Created → Students Enroll → Activities Scheduled → Attendance Recorded → Performance Evaluated → Awards Issued → Reports Generated

3.2 Daily Operational Flow

Student attends club/activity
↓
Coach/teacher records attendance
↓
Performance recorded (if applicable)
↓
Upcoming events scheduled and updated
↓
Notifications sent to students/parents
↓
Participation data updated in student profile

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4. MODULE ARCHITECTURE

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Sports & Co-Curricular Module
│
├── Club & Activity Management
├── Member Enrollment & Management
├── Attendance Tracking
├── Event & Tournament Scheduling
├── Performance Tracking
├── Awards & Recognition
├── Communication & Notifications
├── Analytics & Reporting
├── Integration Layer (Student/Timetable)
├── Mobile & Accessibility
├── Security & Role-Based Access
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5. CLUB & ACTIVITY MANAGEMENT

5.1 Features

- Create and manage clubs, teams, and activities
- Categorize activities (sports, arts, academic clubs, community service)
- Assign teachers/coaches to each activity
- Define schedules and meeting times

5.2 Data Model

Club

- club_id
- name
- category
- coach_id
- schedule

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6. MEMBER ENROLLMENT & MANAGEMENT

6.1 Features

- Student enrollment into clubs and activities
- Manage membership (join/leave/transfer)
- Multi-club participation support
- Track active/inactive members

6.2 Data Model

Membership

- membership_id
- student_id
- club_id
- status
- join_date

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7. ATTENDANCE TRACKING

7.1 Features

- Record attendance for each activity session
- Track participation frequency
- Identify inactive or irregular participants

7.2 Rules

IF attendance below threshold
→ Flag student as inactive
→ Notify coach/parent (optional)

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8. EVENT & TOURNAMENT SCHEDULING

8.1 Features

- Schedule competitions, tournaments, and events
- Define venues, dates, and participants
- Avoid conflicts with academic timetable

8.2 Data Model

Event

- event_id
- name
- type (tournament/meeting/performance)
- date
- venue
- participants

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9. PERFORMANCE TRACKING

9.1 Features

- Record performance metrics (scores, rankings, outcomes)
- Track individual and team performance
- Generate progress reports

9.2 Examples

- Football: Match scores, goals scored
- Debate: Rankings, points
- Music: Performance ratings

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10. AWARDS & RECOGNITION

10.1 Features

- Record and display student achievements
- Categorize awards (sports, arts, academics)
- Highlight top performers

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11. COMMUNICATION & NOTIFICATIONS

11.1 Features

- Announcements for clubs and events
- Notifications for schedule changes
- Alerts for parents and students
- Messaging between coaches and members

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12. ANALYTICS & REPORTING

12.1 Features

- Participation trends (by club, class, gender)
- Performance analytics
- Event attendance reports
- Engagement metrics

12.2 Dashboards

- Activity participation overview
- Performance trends
- Club popularity metrics

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13. INTEGRATION LAYER

13.1 Features

- Link with Student Management for profiles
- Sync with Timetable to avoid scheduling conflicts
- Integrate with Academics for holistic student performance view

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14. MOBILE-FIRST & ACCESSIBILITY

14.1 Requirements

- Mobile-friendly interface for students and coaches
- Push notifications for updates
- Offline support for attendance capture
- Multilingual support (optional)

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15. SECURITY & ROLE-BASED ACCESS

Role	Access Scope
Admin	Full control, reports, configuration
Coach	Manage assigned clubs and students
Student	View activities, join clubs, track progress
Parent	View child's participation and achievements

15.2 Security Features

- Role-based access control
- Data privacy for student records
- Audit logs for activity tracking

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16. SCALABILITY

- Support large numbers of clubs and activities
- Multi-campus support
- Efficient search, filtering, and categorization

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17. IMPLEMENTATION GAPS (CRITICAL FIXES)

17.1 Current Issues Identified

- No member enrollment or participation tracking
- Missing performance metrics and reporting
- No communication tools for clubs
- Weak integration with student and timetable modules
- Limited analytics and scalability features

17.2 Required Fixes

- Implement membership management system
- Add attendance and performance tracking
- Enable communication and notification features
- Integrate with student and timetable systems
- Build analytics dashboards

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18. FUTURE ENHANCEMENTS

- AI-based talent identification
- Integration with national sports/competition systems
- Gamification (badges, leaderboards)
- Video performance analysis tools
- Scholarship and talent tracking